Thaqhiri: Preliminary Systems Engineering Design of an Autonomous UAV for High-Altitude Mine Monitoring

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***Abstract*—Underground mining remains a major source of livelihood in Bolivia while exposing workers to rockfall, toxic gases, low visibility, and structurally unstable tunnels. These risks are amplified in high-altitude mines where reduced air density lowers multirotor thrust margins and where commercial indoor inspection drones may not match local altitude, cost, or maintenance constraints. This paper introduces Thaqhiri, a low-cost autonomous quadrotor design intended for use in mine mapping and for carrying out environmental reconnaissance. The key contribution consists of allocating the mission requirements to a commercial-off-the-shelf architecture through a traceable systems-engineering approach, rather than proposing a new flight-control or SLAM algorithm. The proposed platform combines LiDAR-inertial-visual navigation, onboard mesh generation, and the ability to detect CO, H_2S, O_2, and NO_2, incorporating a local gas overlay to enable pre-entry hazard assessment. Preliminary engineering analysis estimates a 4.123 kg takeoff mass, approximately 12.9–19.6 min of practical hover endurance at 4000 m above sea level, and a thrust-to-weight ratio of 2.03 at nominal 6S voltage for a 17 x 5.8 in propeller configuration. Thaqhiri proposes a worker-focused inspection tool in order to decrease human exposure and help make safer decisions in adverse and hazardous mining enviroments. Thaqhiri has been funded by EPICS in IEEE for physical implementation.**

***Index Terms*—Autonomous UAV, Commercial-Off-The-Shelf (COTS) high-altitude flight, mine safety, systems engineering.**

I. Introduction

The mining industry has for a long time been of great economic, social and cultural significance in Bolivia; zinc, gold, silver, tin and lead comprising a major part of the country's mineral exports in recent decades [1], [2]. At the same time, there are serious safety risks for those working underground. Accounts from the mines in Bolivia mention deaths resulting from exposure to toxic gases, asphyxiation, tunnel collapses and explosions [3], [4]. These problems are especially pertinent in Potosi, which is situated at about 4000 m above sea level, since Cerro Rico has been gradually weakened over the course of centuries while mining operations have continued [5]–[7].

Commercially available inspection drones show the value of carrying out aerial inspections in restricted areas where GNSS signals are not available. For instance, Flyability´s Elios 3 remains a popular choice for mining inspections due to its small size and high resolution mapping. Nevertheless, the maximum altitude specified for the Elios 3 when using the LiDAR equipment is only 3000 m AMSL, which is lower than the elevation of most mining sites in Bolivia. Furthermore, its price point surpassing the 50,000+ USD$ is unaffordable for the economic situation of the global south [8], [9]. Because of these limitations, a different design approach is required: a platform that makes heavy use of COTS components and is maintainable, with a specific focus on high-altitude mine mapping and hazard awareness.

The paper introduces Thaqhiri, an autonomous quadrotor system which is intended for initial reconnaissance and mapping at a design altitude of 4000 meters ASML. The architecture proposed makes use of LiDAR-inertial-visual navigation, onboard mapping, environmental sensing, and autonomous frontier exploration in order to produce spatial information prior to workers entering unexplored areas. The system is to be used as an inspection tool focused on worker safety with a focus on being affordable, easy to maintain locally, and capable of providing accessible digital evidence to support safer decision-making.

The main contribution of this paper is the development of a low-cost autonomous drone concept that would make the use of the quadrotor more accessible to high-altitude mining communities. Rather than competing with specialized commercial inspection platforms, Thaqhiri explores a more affordable design point that can be assembled, maintained, and potentially adapted. The proposed system is evaluated through a requirements and verification baseline, an autonomous frontier-exploration mission concept, and preliminary analyses of mass, propulsion, power, endurance, interfaces, and mission range.

II. Related Work and Design Gap

The safety of underground mines is now all the more reliant on having up-to-date spatial and environmental information. The literature on digital twins suggests that RGB imagery, point clouds, contact sensors, and operational data should be integrated in order to aid monitoring, simulation, anomaly prediction, and decision support [10], [11]. In this regard, a mine map serves not only as a geometric representation but also as a means of showing drift deformation, ventilation conditions, the possibility of gas accumulation, access routes, and rescue limitations. Yet the actual implementation is especially difficult in places like Bolivia due to high initial costs, a lack of specialized personnel, connectivity problems, fragmented

data systems, and the general difficulty of installing reliable underground sensors [10].

Underground inspection robots have to function in situations where Global Navigation Satellite System (GNSS) signals are blocked, with poor lighting, dust, repetitive environments, intermittent communication, and limited energy supplies [12]. Although low-cost sensors can enhance awareness of the environment, factors like calibration, cross-sensitivity, temperature, dust and data integration are still major limitations [13]. As a result, sensor selection should be viewed as a safety-focused multi-criteria decision rather than merely as a bill-of-materials decision [14].

The aircraft Flyability Elios 3 is a commercial benchmark for this type of task. With its basic LiDAR setup, the aircraft has a mass of about 2.412 kg, a maximum flight time of 9.1 minutes, a LiDAR range of 45 m, ± 1.8 cm precision, and a maximum mapping drift of 0.5 percent. It also features LiDAR/VIO-based stabilization, 16,000-lumen lighting, and a collision-resilient protective cage [8]. These specifications must be understood as measurable commercial benchmarks rather than as requirements that Thaqhiri has to meet. The prototype instead has a focus on lower cost, the feasibility of high-altitude propulsion, local maintainability, and humanitarian-use constraints.

The same comparison also shows that Thaqhiri should not be seen as a straightforward copy of an industrial inspection drone. The design has to take into account the propulsion performance at about 4000 m, the need for low-cost sensing, the limited onboard energy, passive environmental monitoring, autonomy, and collision protection. Specifically, gas measurements must not be regarded as certified entry decisions: on a moving multirotor machine, response lag and the airflow caused by the rotors can result in the measured concentration differing from that of the undisturbed air at the calculated position of the vehicle. Thaqhiri therefore regards gas measurements as an advisory layer involving uncertainty rather than as a substitute for certified mine-gas procedures.

Navigation poses a second major challenge. Reliable underground localization usually involves LiDAR-based odometry combined with IMU fusion, scan de-skewing, local pose or factor-graph optimization, and loop closure [15]. For exploration in coal mines under GNSS-denied conditions, LiDAR-visual-inertial methods have also been suggested, visual information being able to supplement LiDAR and inertial measurements when there is adequate lighting and texture [16]. Furthermore, LiDAR-IMU reconstruction focused on mining applications reinforces the value of using fused sensing for three-dimensional tunnel mapping [17]. These findings therefore lead to the adoption of a LiDAR-based SLAM architecture for Thaqhiri, with stereo images and inertial data serving as extra constraints whenever scan matching becomes less reliable.

A. Component Selection and Mission Concept

The proposed mission concept guided the selection of the main hardware components according to the operational requirements of high-altitude underground mine inspection. The Livox Mid-360 was selected as the primary geometric sensing unit for LiDAR-based localization and mapping, while the ZED X Mini stereo camera provides complementary visual information for inspection and perception redundancy. Onboard processing is performed by the ZED Box Mini an NVIDIA Jetson Orin NX carrier board, whereas the Cube Orange+ flight controller is responsible for low-level flight stabilization, inertial sensing, failsafe handling, and actuator control. The propulsion system is designed around four SunnySky X4110S 400-KV brushless motors powered by a 6S 10-Ah Li-ion battery. Together, these components were selected to balance sensing capability, onboard computation, high-altitude propulsion requirements, system integration, local maintainability, and overall platform cost.

TABLE I
PRELIMINARY REQUIREMENTS AND VERIFICATION BASELINE

ID	Requirement	Allocation
R1	Operate at a 4000 m analytical design point.	Propulsion and energy system
R2	Explore unknown reachable space without continuous piloting.	Frontier planner and local occupancy map
R3	Produce a local mesh, trajectory, and gas overlay.	Edge computer and time-stamped logs
R4	Retain 20% battery energy after nominal mission return.	Energy-aware mission manager
R5	Continue LiDAR-inertial navigation when vision degrades.	Multi-modal estimator and health monitor
R6	Sample CO, H_2S, O_2, and NO_2 as advisory channels.	Passive probes, ADC, and MCU
R7	Isolate propellers with a collision cage before field use.	Mechanical subsystem
R8	Exclude explosive-atmosphere operation.	Mission constraints and procedures

The selected perception payload combines a Livox Mid-360 LiDAR, a ZED X Mini stereo camera, and their inertial measurements. The Mid-360 provides 360-degree horizontal geometry, 200,000 points/s, a 265 g mass, and 6.5 W typical power. The visual branch uses the short-baseline ZED X Mini for forward stereo constraints and inspection imagery. A ZED Box Mini with Jetson Orin NX 16 GB provides two native GMSL2 inputs and executes ROS2 autonomy, sensor fusion, frontier selection, mesh generation, and logging.

In this section the proposed system architecture and mission concept are defined, with the perception and computing payload that have been selected being linked to the preliminary requirements baseline shown in Table I. The architecture makes use of LiDAR, stereo vision, inertial sensing and onboard edge computing in order to enable autonomous exploration, mapping and advisory gas monitoring.

B. Estimation, Timing, and Mapping

The navigation design separates the autonomy estimate from the Cube's flight-control estimate. Let the autonomy state contain position, velocity, attitude, and inertial biases, $\mathbf{x} = [\mathbf{p}, \mathbf{v}, \mathbf{q}, \mathbf{b}_a, \mathbf{b}_g]$. The Mid-360 IMU propagates this state at high rate. LiDAR scan-matching factors provide the primary geometric correction, while visual-inertial factors are enabled only when image exposure, feature count, and residual consistency satisfy health gates.

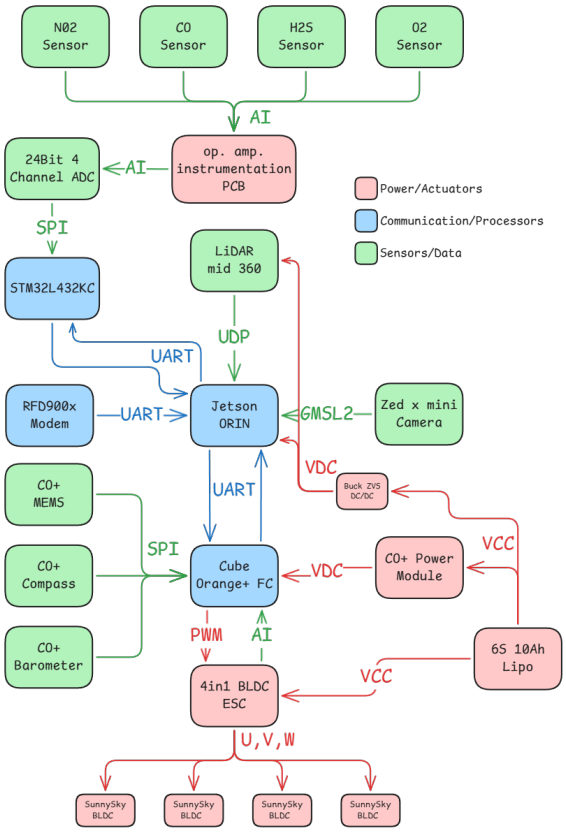


Fig. 1. Thaqhiri Hardware Architecture

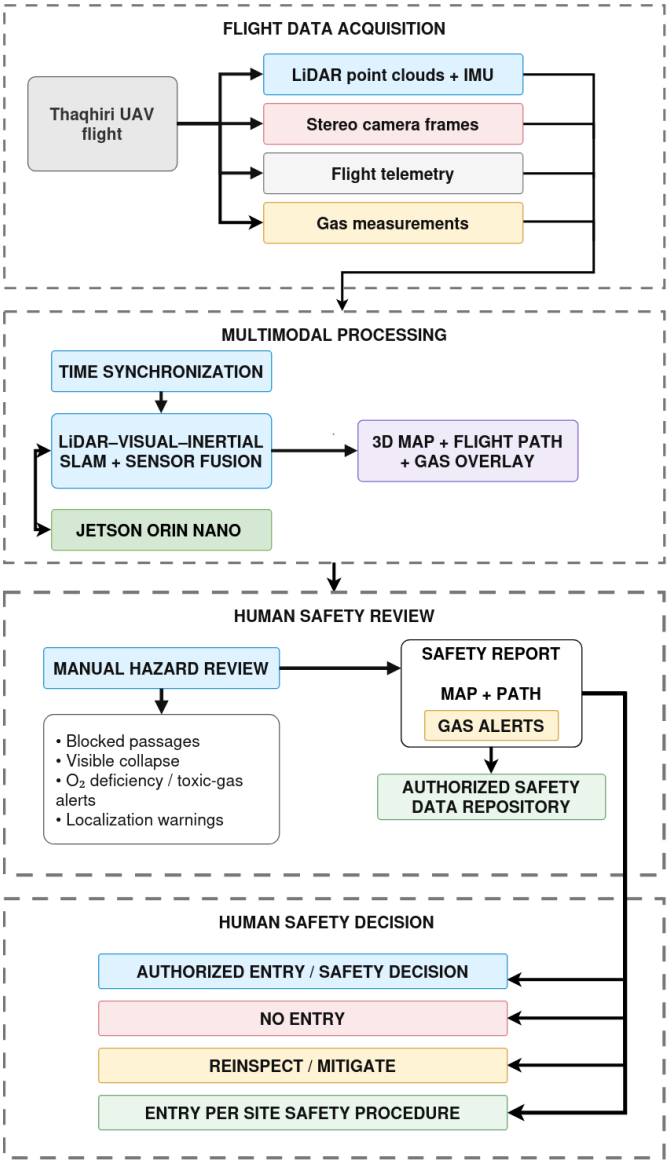


Fig. 2. System Workflow Diagram

The design requires calibrated rigid transforms among the LiDAR, camera, IMU, and vehicle body. Sensor receipt time is insufficient for aggressive motion: the Mid-360 supports PTP synchronization, whereas the camera, gas MCU, and Cube expose different clock domains. The implementation must estimate or discipline their clock offsets and record a synchronization-health flag with every mission. Without that step, geometric fusion and gas-to-pose association can be biased even when each sensor is individually accurate.

C. Interface and Data Budget

Sensor compatibility spans electrical, protocol, timing, and throughput constraints. When the data rate is 200,000 points per second and the size per point is 16 bytes, LiDAR logging demands a rate of 3.2 MB per second (amounting to 3.1 GB over a period of 16 minutes), whereas two streams with a resolution of 1920×1200 and 8 bits per pixel at 60 frames/s require approximately 276 MB/s (more than 265 GB). Consequently, 256 GB of storage is not sufficient to allow continuous raw stereo recording and therefore it is necessary to use hardware encoding, reduce frame rate, insert keyframes, or exploring image compression. The Cube Orange+ is responsible for low-level stabilization and control of the actuators, while the autonomy estimator supplies, through external navigation, time-aligned pose, velocity, covariance values and reset counters; the edge computer transmits bounded position and velocity setpoints, not direct motor commands. In the event that this feed is lost, the vehicle can only rely on IMU-based attitude and barometric altitude stabilization prior to carrying out the landing procedure. Vision is employed whenever there is adequate illumination and texture; if the visual system's condition degrades, LiDAR-inertial navigation is activated, the speed is reduced, or the frontier is rejected. The return operation starts early so that 20% of conservative usable capacity is preserved; the telemetry includes health and mission data while the complete logs stay on board.

III. Analytical Design Verification

A. Mass and Altitude Sizing

II denotes the weight and monetary cost of the chosen components, the resulting 4.12385 kg value is a mass target based on a provisional 1.376 kg integration allowance, not a

TABLE II
PRELIMINARY FLIGHT-UNIT MASS AND COST TARGET BUDGET

Item	Mass (g)	Cost (USD)
ZED X Mini, cable, and mounting allowance	150.00	614.00
Livox Mid-360	265.00	1000.00
Cube Orange+	73.00	405.00
ZED Box Mini NX	700.00	1649.00
Airborne telemetry and antenna allowance	59.85	300.00
6S 10 Ah Li-ion battery	840.00	338.00
Four SunnySky X4110S motors	660.00	188.00
Structure & electronics allowance	1376.00	830.00
Provisional single-flight-unit total	**4123.85**	**5324.00**

closed physical estimate. Using that target and four rotors, the required hover thrust is

$$T_h = \frac{mg}{4} = 10.11 \text{ N} = 1031 \text{ gf/motor}. \quad (1)$$

At 4000 m, using $\rho = 0.82$ kg/m^3 and $\rho_0 = 1.225$ kg/m^3, the density ratio is $\sigma = 0.6694$. With $T \propto \rho n^2 D^4$, the equivalent sea-level table point is

$$T_{eq} = \frac{T_h}{\sigma} = 1540 \text{ gf/motor}. \quad (2)$$

Interpolation of the published X4110S 400 KV, 6S, 17×5.8 in data gives $P_{eq} = 187.9$ W and $I_{eq} = 8.45$ A per motor [18], [19]. These are equivalent table values, not measured high-altitude consumption.

The manufacturer maximum is 3130 gf at 22.2 V. Simple fixed-RPM density scaling gives a propulsion-only thrust-to-weight screening estimate of 2.03. The 25 V table point gives 2.34, but it is only a proxy for a charged 6S pack. At fixed voltage, reduced aerodynamic torque changes motor speed, current, and efficiency; propeller Reynolds effects and cage interference are also omitted. Battery, connector, distribution, thermal, and voltage-sag limits can further reduce both values. Full-throttle system capability is therefore not claimed [20].

The governing sensitivities also expose why the unverified structural allowance matters. Since $T_{eq} \propto m/\rho$ and ideal induced power satisfies $P_i \propto m^{3/2}\rho^{-1/2}$,

$$\frac{\Delta T_{eq}}{T_{eq}} \approx \frac{\Delta m}{m} - \frac{\Delta \rho}{\rho}, \qquad \frac{\Delta P_i}{P_i} \approx \frac{3}{2}\frac{\Delta m}{m} - \frac{1}{2}\frac{\Delta \rho}{\rho}. \quad (3)$$

A 10% mass increase therefore raises ideal hover power by approximately 15% before profile, electrical, or cage-interference losses. Reporting extra mass only as reduced endurance would understate its simultaneous effect on thrust margin and current-path loading.

B. Power and Endurance Sensitivity Scenarios

For total rotor area $A = 0.586$ m^2, momentum theory gives disk loading $W/A = 69.0$ N/m^2 and ideal induced hover power

$$P_i = \frac{W^{3/2}}{\sqrt{2\rho A}} = 262 \text{ W}. \quad (4)$$

Scaling all table electrical power by σ assumes that copper, iron, ESC, and fixed losses follow aerodynamic density, while retaining the full table value assumes none of the load-dependent input decreases. Neither is a rigorous bound. They are retained as sensitivity cases: $4\sigma P_{eq} = 503$ W and $4P_{eq} = 751$ W. A third scenario retains the sea-level non-induced residual and gives 622 W, but it also requires calibration before being treated as predictive.

TABLE III
ASSUMED HOVER-POWER AND ENDURANCE SCENARIOS

Scenario	Prop. (W)	Total (W)	Time (min)
Density-scaled case	503	543	19.6
Residual-loss case	622	667	16.0
Unscaled-table case	751	824	12.9

The listing-based battery allocation is 222 Wh nominal and 177.6 Wh at an assumed 80% usable-energy fraction. Table III adds 40, 45, and 73 W avionics cases. The first two are planning allocations; the upper case uses the ZED Box supply ceiling plus LiDAR, flight-controller, and telemetry loads. None is a measured workload. Exact gas-board, conversion, cage, and installation effects remain unresolved.

These scenario-based values are calculated hover durations, not bounds or mission endurance. Maneuvering, wall and ceiling effects, battery aging, temperature, cage interference, and the final payload can shorten them.

C. Electrical Interface Compatibility

The power budget is not closed merely because the motor can produce the required thrust. For the 667 W scenario, idealized pack current ranges from 26.5 A at 25.2 V to 41.2 A at the listing's 16.2 V discharge floor, before voltage sag. It is 30.0 A at nominal voltage, already equal to the published continuous rating of the stock PM07 XT60/harness even though the PM07 PCB itself has a higher rating [21]. The minimum under-load voltage that preserves hover and control margin must be set well above an absolute cell floor. At the motor's published sea-level maximum point, four channels request 110.8 A before avionics; this is a table screening value, not predicted 4000 m current, and it exceeds the battery listing's inconsistent 80 A burst statement.

The preliminary architecture therefore requires three changes before detailed design release: a propulsion distribution path and connector set rated above the credible system limit; a regulator whose continuous and transient rating covers the ZED Box, camera, and cooling load; and separately protected compute, sensing, and flight-control rails. The selected 4-in-1 ESC is electrically oversized per motor but was designed for smaller FPV platforms, so aggregate thermal performance with four large slow-turning motors remains a verification item [22]. Likewise, the Pololu converter is acceptable only if measured compute load remains below its input- and temperature-dependent limit; its headline 4.5 A value is not a universal guarantee at 22.2 V input [23].

D. Gas-Measurement Allocation

The architecture reserves four passive channels for CO, H_2S, O_2, and NO_2. Exact probe models remain open; con-

TABLE IV
DESIGN-GAP COMPARISON WITH THE ELIOS 3 BENCHMARK

Attribute	Thaqhiri preliminary design	Elios 3
Mass	4.12 kg; integration TBD	2.412 kg with LiDAR
Flight time	12.9–19.6 min assumed hover	9.1 min max. with LiDAR
Altitude	4000 m analytical; unqualified	+3000 m AMSL payload operation
Mapping	Local mesh and trajectory; accuracy TBD	45 m range; ±1.8 cm; 0.5% max. drift
Autonomy	Frontier exploration; energy-aware return	Pilot-operated; LiDAR/VIO; auto-return
Gas	Toxic-gas/O_2 advisory channels; TBD	Flammable-gas/LEL warning; < 20 s response
Protection	Cage required; performance TBD	Carbon cage; collision/fall limits published
Cost	∼$5324 selected components; incomplete	$> \$50,000$ Premium LiDAR package
Maintainability	COTS; university-serviceable; no metric	Commercial service, training, global support

sequently, no accuracy, calibration, or alarm-threshold claim is made. A "24-bit ADC" also does not establish 24-bit system accuracy. Each electrochemical cell requires a sensor-specific bias or load, low-bias-current transimpedance stage, anti-alias filtering, reference, and fault detection before simultaneous conversion by the ADS131A04 and acquisition by the STM32L432KC [24], [25]. For a current-output cell with sensitivity S_i and feedback resistance R_f, the provisional conversion is

$$\hat{C}_i = \frac{V_i - V_{0,i}}{S_i R_f}, \tag{5}$$

where S_i is defined as a signed current sensitivity that includes analog-front-end polarity. Its value, zero offset, cross-sensitivity, and pressure/temperature compensation must come from the exact probe and its calibration lot. In particular, an O_2 cell may respond to oxygen partial pressure; a sea-level calibration cannot be assumed valid at 4000 m simply because ambient oxygen remains near 20.9% by volume.

For a first-order probe,

$$\tau \dot{y} + y = C(t), \qquad t_{90} = 2.303\tau, \tag{6}$$

so motion at speed v creates a nominal lag distance $d_{90} = vt_{90}$. A useful map representation attaches separate location and concentration uncertainties to a region rather than a point. Provisional first-order models are

$$\Sigma_x \approx \Sigma_p + \mathbf{v}\mathbf{v}^T \sigma_\tau^2 + \Sigma_{flow,x}, \tag{7}$$

and

$$\sigma_C^2 \approx \sigma_{sensor}^2 + \sigma_{cal}^2 + \sigma_{cross}^2 + \sigma_{flow,C}^2. \tag{8}$$

The first is a 3D sample-location covariance; the second is concentration variance. Both must be calibrated for the installed probe and airflow. Frontier utility can favor a stationary dwell when a sustained concentration change is observed, but hovering does not remove propwash. Gas samples must therefore be displayed as delayed uncertainty regions and never as proof that a mapped location is safe.

IV. COMMERCIAL BENCHMARK AND LIMITATIONS

A. Prototype Component Cost Structure

The component estimate is built from one flight-unit quantity rather than the procurement BOM's development spares. Table II totals approximately USD 5324, excluding taxes, contingency, ground support, unresolved cage and integration hardware, calibration equipment, and engineering labor. It is therefore a prototype component budget, not a product purchase price or life-cycle cost.

B. Commercial Gap Analysis

Table IV compares design intent with published Elios 3 performance. It is a gap analysis, not an equivalence claim. The public Elios figure refers to a USD 50,000+ Premium LiDAR package, not a bare-aircraft price [26]. The large difference includes product maturity, software, support, training, and verified protection that are absent from the Thaqhiri estimate.

V. CONCLUSION

Thaqhiri is presented as a preliminary autonomous mine-inspection system design, not as a validated robot or a new autonomy algorithm. Its contribution is the traceable co-design of high-altitude propulsion, LiDAR-inertial-visual navigation, frontier exploration, local mesh generation, and diffusion-fed environmental sensing around a COTS architecture. With a provisional 4.12 kg mass target, the 4000 m analysis gives a 2.03 fixed-RPM motor-propeller screening estimate and 12.9–19.6 min across three assumed hover-power scenarios. The comparison with Elios 3 frames altitude-oriented sizing, component accessibility, and lower selected-component cost as design objectives, not demonstrated advantages. It also exposes the larger mass and unresolved protection, power, sensing, landing, and verification gaps. The next design phase must close the power-distribution and mechanical baselines before controlled subsystem and integrated prototype verification.

VI. FUTURE WORK

The immediate next step is software-in-the-loop (SIL) verification in a physics-based simulation environment. The simulated vehicle will be parameterized using the selected hardware and the preliminary physical properties of the current design, including mass and inertia, reduced air density at 4000 m AMSL, motor–propeller performance, battery voltage sag, actuator limits, collision-cage interference, sensor noise, communication delays, and onboard computing constraints. These experiments will evaluate localization, frontier exploration, energy-aware return, degraded-sensor behavior, and failsafe transitions under representative tunnel geometries, lighting conditions, and communication loss. Gas-sensing simulations will also account for probe response time and the spatial uncertainty introduced by vehicle motion and propeller-induced

airflow. Nevertheless, SIL results will remain limited by the fidelity of these models and will not constitute validation of the physical aircraft.

Following SIL testing, potential hardware-in-the-loop (HIL) validation will connect the Cube Orange+ flight controller and onboard edge computer to the simulated vehicle and sensor environment. This stage will support verification of interfaces, timing, external-navigation messages, command limits, estimator resets, and failsafe behavior using the intended computing hardware. Subsequent work will include physical assembly, bench and tethered testing, high-altitude flight characterization, sensor calibration, and progressively controlled underground trials. Subject to site authorization and applicable safety procedures, final validation is intended to take place in operational environments inside Bolivian mines. The authors gratefully acknowledge EPICS in IEEE for funding the Thaqhiri project with a 7,600 USD grant and supporting its forthcoming physical assembly and validation in real underground mining environments in Bolivia.

REFERENCES

[1] F. R. Lopez, G. G. Prudencio, B. de Marchi Moyano, and D. E. Pacheco, "Efectos de la mineria en el desarrollo economico, social y ambiental del estado plurinacional de bolivia," United Nations Economic Commission for Latin America and the Caribbean, Tech. Rep., 2020, accessed: Aug. 30, 2026. [Online]. Available: https://minsus.net/mineria-sustentable/wp-content/uploads/2020/06/PDF-Efectos-de-mineria-en-desarrollo-de-Bolivia.pdf

[2] A. C. Sarmiento and P. B. Anaya, "Mineria en bolivia: Implicancias en el comercio y produccion," Banco Central de Bolivia, Tech. Rep., 2019, reviewed by Jose Antonio Caballero. Accessed: Aug. 30, 2026. [Online]. Available: https://www.bcb.gob.bo/webdocs/publicacionesbcb/2021/05/52/40%20-%20Mineria%20en%20Bolivia%20Implicancias%20en%20el%20comercio%20y%20produccin.pdf

[3] Brujula Digital. (2024) Mineria ilegal en bolivia: 119 muertes en lo que va del 2024, casi 11 obreros por mes. Accessed in project proposal May. 25, 2026. [Online]. Available: https://brujuladigital.net/economia/2024/12/03/mineria-ilegal-en-bolivia-119-muertes-en-lo-que-va-del-2024-casi-11-obreros-por-mes-40454

[4] Radio Pio XII. (2026) Mueren 425 mineros en accidentes en ultimos 4 anos en minas de bolivia. Accessed: May. 20, 2026. [Online]. Available: https://radiopio12.com.bo/mueren-425-mineros-en-accidentes-en-ultimos-4-anos-en-minas-de-bolivia/

[5] Ministerio de Mineria y Metalurgia, "Situación de la minería y boletín estadístico al trecer trimestre 2025," Ministerio de Mineria y Metalurgia, Bolivia, Tech. Rep., 2025, accessed: Aug. 30, 2026. [Online]. Available: https://www.mineria.gob.bo/wp-content/uploads/2026/01/Boletin_3T-2025.pdf

[6] Y. M. Cayo. (2025) El cerro rico absorbio la vida de casi 100 mineros este ano y su cuspide esta por colapsar. Accessed: Jun. 15, 2026. [Online]. Available: https://eldeber.com.bo/pais/el-cerro-rico-absorbio-la-vida-de-casi-100-mineros-este-ano-y-su-cuspide-esta-por-colapsar_530776/

[7] E. Potosi. (2025) Reportan que alrededor de 30 mil personas trabajan en el cerro rico. Accessed: May. 17, 2026. [Online]. Available: https://elpotosi.net/local/20250914_reportan-que-alrededor-de-30-mil-personas-trabajan-en-el-cerro-rico.html

[8] Flyability, *Elios 3 Technical Documentation*, 2026, accessed 24 June 2026. [Online]. Available: https://3798671238-files.gitbook.io/~/files/v0/b/gitbook-x-prod.appspot.com/o/spaces%2FEUlQdNranSJ0ddu4qSxa%2Fuploads%2F9gRokOPsGVhfPZz4y88R%2FElios%203%20Technical%20Specifications%20v1.1.pdf?alt=media&token=948d790d-751a-42e7-a038-cd582bab857d

[9] Flyability. (2022) Mining drones: Inspect, survey, and map inaccessible areas from a safe distance. [Online]. Available: https://www.flyability.com/mining-drones

[10] C. Cacciuttolo, E. Atencio, S. Komarizadehasl, and J. A. Lozano-Galant, "Development of an advanced multi-layer digital twin conceptual framework for underground mining," *Sensors*, vol. 25, no. 21, p. 6650, 2025. [Online]. Available: https://www.mdpi.com/1424-8220/25/21/6650

[11] H. Wen, S. Liu, X. Zheng, G. Cai, B. Zhou, W. Ding, and Y. Ma, "The digital twins for mine site rescue environment: Application framework and key technologies," *Process Safety and Environmental Protection*, vol. 186, pp. 176–188, 2024.

[12] M. Konieczna-Fulawka, A. Koval, G. Nikolakopoulos, M. Fumagalli, L. Santas Moreu, V. Vigara-Puche, J. Muller, and M. Prenner, "Autonomous mobile inspection robots in deep underground mining–the current state of the art and future perspectives," *Sensors*, vol. 25, no. 12, p. 3598, 2025.

[13] C. Cacciuttolo, V. Guzman, P. Catrinir, E. Atencio, S. Komarizadehasl, and J. A. Lozano-Galant, "Low-cost sensors technologies for monitoring sustainability and safety issues in mining activities: Advances, gaps, and future directions in the digitalization for smart mining," *Sensors*, vol. 23, no. 15, p. 6846, 2023.

[14] Q. Wang, T. Cheng, Y. Lu, H. Liu, R. Zhang, and J. Huang, "Underground mine safety and health: A hybrid MEREC-CoCoSo system for the selection of best sensor," *Sensors*, vol. 24, no. 4, p. 1285, 2024.

[15] K. Ebadi, L. Bernreiter, H. Biggie, G. Catt *et al.*, "Present and future of SLAM in extreme environments: The DARPA SubT challenge," *IEEE Transactions on Robotics*, vol. 40, pp. 936–959, 2024.

[16] S. Wang, Y. Liang, S. Liao, S. Xiao, and L. Wang, "Multi-sensor information fusion methods for coal mine exploration in gnss-denied scenarios," *EURASIP Journal on Wireless Communications and Networking*, vol. 2026, no. 4, 2025. [Online]. Available: https://doi.org/10.1186/s13638-025-02490-9

[17] Q. Ma, J. Chen, Y. Chen, X. Xu, and C. Wang, "A three-dimensional reconstruction method of coal mine tunnel fused with lidar and imu," *Coal Science and Technology*, vol. 53, no. 2, pp. 351–362, 2025. [Online]. Available: https://www.mtkxjs.com.cn/en/article/doi/10.12438/cst.2024-1386

[18] SunnySky USA, "Sunnysky XS high power X4110S brushless motors," 2026, accessed: Aug. 31, 2026. [Online]. Available: https://sunnyskyusa.com/products/sunnysky-x4110s-motors?srsltid=AfmBOooMuAYTXCTbf5Nhg8XnxZzyt4bUslqWxpwKEtsSPszcBj-E6pfx

[19] J. B. Brandt and M. S. Selig, "Propeller performance data at low reynolds numbers," in *49th AIAA Aerospace Sciences Meeting Including the New Horizons Forum and Aerospace Exposition*, no. AIAA 2011-1255, 2011.

[20] Lumenier.com, "Upgrade energy green 10000mah 6S 20C li-ion battery - XT60," 2026, accessed: Aug. 31, 2026. [Online]. Available: https://www.lumenier.com/products/upgrade-energy-green-10000mah-6s-20c-li-ion-battery-xt60

[21] Holybro, "PM07 Power Module (14S): Quick Start Guide," Product technical documentation, 2025, accessed: Aug. 31, 2026. [Online]. Available: https://docs.holybro.com/power-module-and-pdb/power-module/pm07-quick-start-guide

[22] HOBBYWING, "XRotor FPV G2 65A 4in1 ESC - FPV drone competition ESC," 2026, accessed: Aug. 31, 2026. [Online]. Available: https://www.hobbywing.com/en/products/xrotor-fpv-g2-65a-4in1-esc268

[23] Pololu Corporation, "12 V, 4.5 A Step-Down Voltage Regulator D36V50F12," Product specifications, 2026, accessed: Aug. 31, 2026. [Online]. Available: https://www.pololu.com/product/4095

[24] Texas Instruments, *ADS131A0x 2- or 4-Channel, 24-Bit, 128-kSPS, Simultaneous-Sampling, Delta-Sigma ADC*, revision e ed., Texas Instruments, 2020, accessed: Aug. 31, 2026. [Online]. Available: https://www.ti.com/lit/ds/symlink/ads131a04.pdf

[25] STMicroelectronics, *STM32L432KC Ultra-Low-Power Arm Cortex-M4 32-Bit MCU*, STMicroelectronics, 2026, accessed: Aug. 31, 2026. [Online]. Available: https://www.st.com/en/microcontrollers-microprocessors/stm32l432kc.html

[26] X. Su, R. Chen, J. Ma, C. Li, and J. E. Froehlich, "Flymethrough: Human-ai collaborative 3d indoor mapping with commodity drones," 2025. [Online]. Available: https://arxiv.org/abs/2508.20034